Practice workbook · Calculate an amount from a percentage, fraction or probability

A total population, volume or mass is given with a percentage, fraction or probability; the matching number, or the new value after a percentage rise or fall, is wanted.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-AMOUNT-FROM-PERCENTAGE

1. Question 7 · part c.i · 2 marks

June 2022 Paper 1

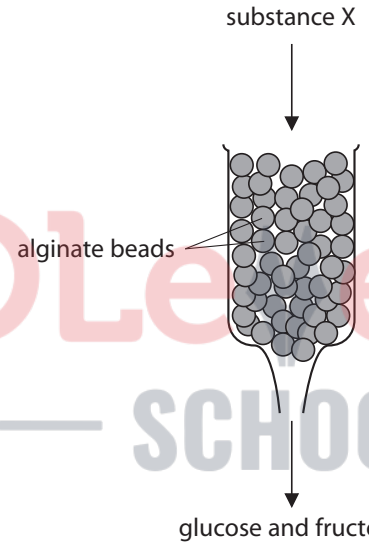
Worked-example retry · Attempt part c.i only.

7 (a) Industrial processes can use immobilised enzymes.

(i) Give one reason why using immobilised enzymes can cost less money than using enzymes that are not immobilised.

(1)

(ii) The diagram shows the apparatus used to produce glucose and fructose using immobilised enzymes attached to alginate beads.



Name substance X.

(1)

(iii) Describe how to prepare alginate beads.

(3)

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1. Question 7 · part c.i · 2 marks

June 2022 Paper 1

Worked-example retry · Attempt part c.i only.

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(b) Which process is involved in the conversion of glucose into glycogen in the human body?

(1)

- ☒ **A** glucagon is released from the pancreas and travels to the liver
- ☒ **B** insulin is released from the pancreas and travels to the liver
- ☒ **C** glucagon is released from the liver and travels to the pancreas
- ☒ **D** insulin is released from the liver and travels to the pancreas

(c) (i) The total population of the United Kingdom is 66.65 million.

Diabetes affects 6.0% of this population.

Calculate the number of people with diabetes in the United Kingdom.

(2)

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number of people = ... million

(ii) Surveys were carried out in different areas of the United Kingdom.

These surveys collected data on the number of people in each area with diabetes.

The information collected is shown in the box.

38	62	51	96	49	78	82	91	65

Determine the median number of people with diabetes.

(2)

median = ...

(Total for Question 7 = 10 marks)

1. Question 7 · part c.i · 2 marks · Working and answer

Working and answer

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2. Question 4 · part a · 2 marks

June 2019 Paper 2

Further practice · Attempt part a only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

In 1916, an American doctor, Joseph Goldberger, noticed symptoms of a mystery disease that were common among his patients. These symptoms included headaches, swollen tongues, skin rashes, upset stomachs and mental illness. In one American state, North Carolina, 40% of the 30 000 people living in this area died from this mystery disease.

Goldberger investigated whether the symptoms of the mystery disease were caused by pathogens. His investigation involved injecting himself with the blood from one patient, eating the skin rash of another and swallowing fluid taken from the intestines of a third. Goldberger did not get the disease, although he did suffer from nausea and diarrhoea.

Goldberger concluded that the symptoms shown by the patients were not caused by an infectious disease. He decided that they were more likely to be caused by the poor diet of the patients.

(a) Calculate the number of people in North Carolina who died from this mystery disease (lines 4 to 5).

(2)

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number of deaths = ...

(b) State the meaning of the term pathogens (line 7).

(1)

.....

.....

(c) The people were suffering from a deficiency disease called pellagra, which is caused by a lack of vitamin B in the diet.

State why deficiency diseases such as pellagra are not infectious diseases.

(1)

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2. Question 4 · part a · 2 marks · Working and answer

Working and answer

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3. Question 4 · part c.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part c.ii only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Obesity and diabetes have been linked to a 20% increase in the number of leg and foot amputations.

Data from Diabetes UK shows that there were 26 378 lower limb amputations related to diabetes from the period 2014 to 2017. This is an increase from the period 2010 to 2013 when there were 22 092 amputations related to diabetes.

Minor lower limb amputations are below the ankle and major lower limb amputations are below the knee. Minor lower limb amputations increased by 26.5% and major lower limb amputations increased by 4.0%.

In 2019 in the United Kingdom 3.8 million people had been diagnosed with diabetes and 90% of these people had Type 2 diabetes.

(a) (i) Describe the causes of obesity.

(3)

(ii) Give two health risks, other than diabetes, associated with obesity.

(2)

1 .....

2 .....

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3. Question 4 · part c.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part c.ii only.

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(b) Diabetes is caused as a result of insulin not being secreted or not working correctly.

(i) Which organ secretes insulin?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

(ii) Insulin stimulates the conversion of glucose into glycogen.

In which organ does this occur?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

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(c) (i) Calculate the percentage increase in the number of lower limb amputations between the period 2010 to 2013 and the period 2014 to 2017.

(3)

percentage increase = ... %

(ii) Calculate the number of people in 2019 who had been diagnosed with Type 2 diabetes in the United Kingdom.

(1)

number of people = ... million

3. Question 4 · part c.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part c.ii only.

(d) Diabetes causes a narrowing of the arteries supplying blood to the lower limbs.

Suggest why this can lead to the need to amputate the lower limbs.

(3)

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(Total for Question 4 = 14 marks)

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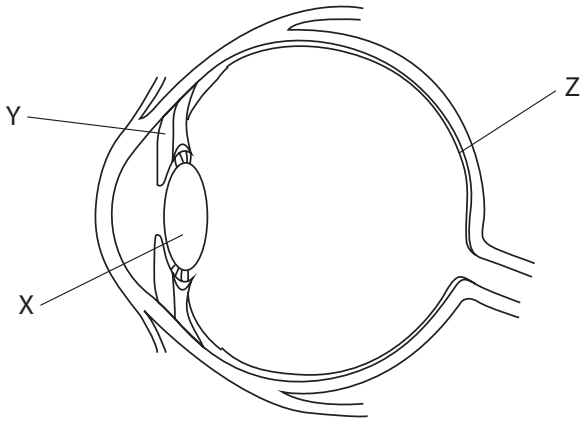
Working and answer

4. Question 1 · part b.iii · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.iii only.

1 (a) The diagram shows a human eye.



(i) Draw one line from each structure to its correct function.

(3)

Structure	Function
	transmits electrical impulses to the brain
structure X	refracts light rays
	protects the eye from pathogens
structure Y	
	detects light rays
structure Z	controls the shape of the lens
	controls the size of the pupil

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4. Question 1 · part b.iii · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.iii only.

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(ii) Most humans have stereoscopic vision.

Which of these statements occurs because of stereoscopic vision?

(1)

- ☒ **A** being able to judge depth
- ☒ **B** being able to see in colour
- ☒ **C** having a wide field of view
- ☒ **D** to be able to see near and far objects

(b) The diagram shows a healthy eye and an eye with a cataract.



Healthy eye Eye with cataract

(i) Describe the cause of cataracts.

(2)

.....

.....

.....

(ii) State one treatment for cataracts.

(1)

.....

(iii) It is estimated that 39 million people in the world are blind.

Of these people, 43% are blind because of cataracts.

Determine the estimated number of people who are blind because of cataracts.

(2)

number of people = ...

(iv) Some people are blind because of a vitamin deficiency.

Name the vitamin that is needed to prevent this type of blindness.

(1)

.....

(Total for Question 1 = 10 marks)

4. Question 1 · part b.iii · 2 marks · Working and answer

Working and answer

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5. Question 5 · part b.ii · 3 marks

January 2022 Paper 1

Further practice · Attempt part b.ii only.

5 (a) Describe the possible effects of a DNA mutation.

(3)

(b) Changes to whole chromosomes can happen when meiosis does not occur properly.

These changes are called chromosome aberrations.

The table shows the probability of children being born with a chromosome aberration to mothers of different ages.

Age of mother in years	15–19	20–24	25–29	30–34	35–39	40–44
Probability of chromosome aberration	0.0010	0.0010	0.0010	0.0015	0.0045	0.0140

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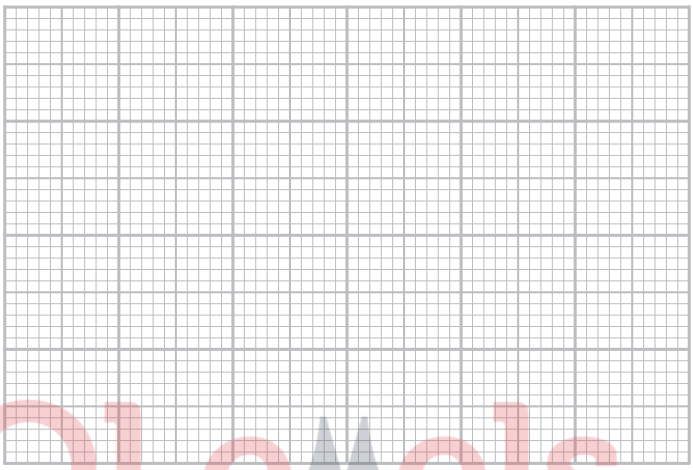
5. Question 5 · part b.ii · 3 marks

January 2022 Paper 1

Further practice · Attempt part b.ii only.

(i) Plot a bar chart of the results.

(5)



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(ii) In one year, 43 000 children are born to mothers aged 30–34.

Calculate the probable number of these children who will have a chromosome aberration.

Give your answer to two significant figures.

(3)

number of children = ...

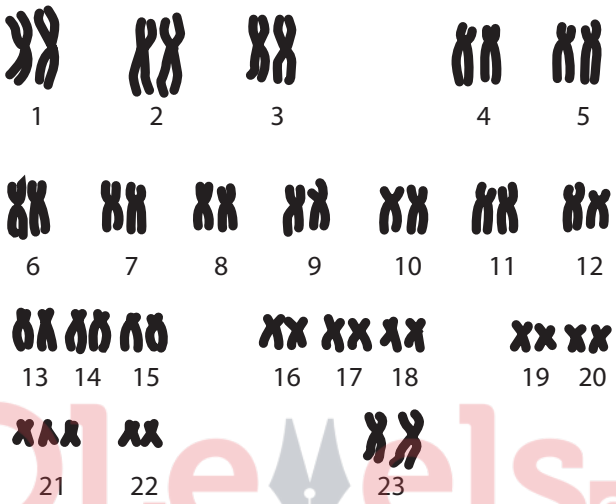
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5. Question 5 · part b.ii · 3 marks

January 2022 Paper 1

Further practice · Attempt part b.ii only.

(c) The diagram shows the full diploid number of chromosomes in a person.



Justify the statement that this diagram shows the genotype of a female with a chromosome aberration.

(4)

(Total for Question 5 = 15 marks)

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5. Question 5 · part b.ii · 3 marks · Working and answer

Working and answer

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6. Question 6 · part a.i · 3 marks

June 2025 Paper 1R

Further practice · Attempt part a.i only.

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6 (a) Red-green colour blindness is caused by a genetic mutation.

The table shows the percentage of red-green colour blind males and colour blind females in different regions.

Equal numbers of males and females were tested in each region.

Region	Total number of people tested	Percentage of colour blind males	Percentage of colour blind females
A	150	10.00	1.00
B	16 000	6.60	0.40
C	5000	0.80	0.01

(i) Calculate the number of colour blind males in the test group in region C.

(3)

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number of males = ...

(ii) State why the results from region A are not considered to be reliable.

(1)

.....

6. Question 6 · part a.i · 3 marks · Working and answer

Working and answer

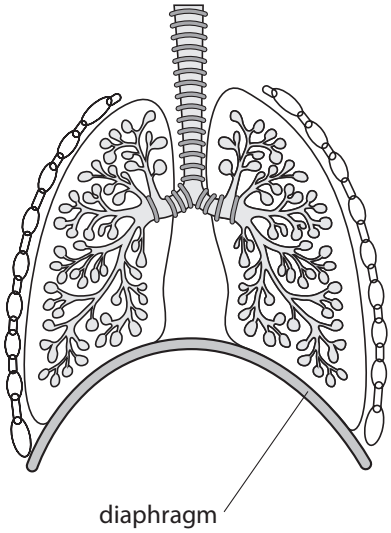
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7. Question 4 · part b.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part b.ii only.

4 (a) The diagram shows the human lungs.



(i) Add the following labels on the diagram

(3)

- trachea
- bronchus
- alveoli

(ii) Give four features of the lungs that adapt them for gas exchange.

(4)

1 .....

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2 .....

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3 .....

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4 .....

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7. Question 4 · part b.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part b.ii only.

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(b) A spirometer can be used to test lung function.

(i) Describe how a spirometer is used to obtain values for tidal volume.

(4)

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(ii) Asthma is a condition that causes the tubes in the lungs to narrow.

A person without asthma has a vital capacity of 4600 cm^3.

People with severe asthma have a 30% lower vital capacity compared with a healthy person.

Calculate the vital capacity of a person with severe asthma.

(2)

vital capacity of a person with severe asthma = ... cm^3

(Total for Question 4 = 13 marks)

7. Question 4 · part b.ii · 2 marks · Working and answer

Working and answer

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8. Question 5 · part b.ii · 2 marks

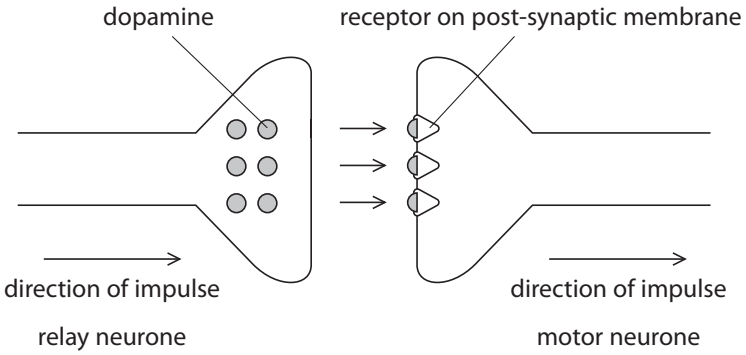
June 2023 Paper 2R

Further practice · Attempt part b.ii only.

5 (a) Dopamine is a neurotransmitter produced by nerve cells in the brain.

The nerve cells that produce dopamine are involved in stimulating movement and increasing heart rate.

The diagram shows how dopamine travels from a relay neurone to a motor neurone in the brain.



(i) Describe what is meant by a neurotransmitter.

(2)

(ii) Describe how information is transmitted from the relay neurone to the motor neurone.

(2)

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8. Question 5 · part b.ii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part b.ii only.

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(iii) Suggest how an excess of dopamine will affect bodily functions.

(2)

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(b) In 2017 in the United Kingdom, 145 000 people were diagnosed with Parkinson's disease.

This figure is expected to have an 18% increase by 2025.

(i) Give two symptoms of Parkinson's disease.

(2)

1 .....

.....

2 .....

.....

(ii) Calculate the number of people expected to be diagnosed with Parkinson's disease in 2025.

(2)

number of people = ...

(Total for Question 5 = 10 marks)

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8. Question 5 · part b.ii · 2 marks · Working and answer

Working and answer

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9. Question 4 · part a.iii · 3 marks

June 2025 Paper 2

Further practice · Attempt part a.iii only.

4 A student investigates the effect of different concentrations of sugar solution on the mass of cylinders made from potato.

This is the student's method.

- push a cork borer into different parts of a potato
- remove five cylinders of potato
- wrap each cylinder in a piece of filter paper
- remove from the filter paper and find the mass of each cylinder
- place one cylinder into distilled water
- place each of the other cylinders into different concentrations of sugar solution
- remove the cylinders after two hours and wrap in filter paper
- remove from the filter paper and find the new mass

These are the student's results.

1.00 au = −7 %	1.50 au = −10 %	0.00 au (distilled water) = +8 %
	0.25 au = +4 %	0.50 au = −1 %

The student's results are given as the percentage change in mass of the cylinder and the concentration of sugar is given in arbitrary units (au).

(a) (i) Draw a table to display the results.

(4)

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9. Question 4 · part a.iii · 3 marks

June 2025 Paper 2

Further practice · Attempt part a.iii only.

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(ii) Determine the median value for sugar concentration.

(1)

median value = ...

(iii) The original mass of the potato cylinder placed into the sugar solution, of concentration 0.25 au, was 7.0 g.

Calculate the mass of the cylinder after two hours.

Give your answer to two significant figures.

(3)

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mass of cylinder = ... g

(b) Use the student's results to estimate the concentration of sugar solution that would produce no change in the mass of the potato cylinder.

You should include details of how you obtained your answer.

(4)

concentration of sugar solution = ... au

9. Question 4 · part a.iii · 3 marks · Working and answer

Working and answer

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10. Question 2 · part c · 3 marks

January 2022 Paper 2

Further practice · Attempt part c only.

2 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The breathing system consists of airways that allow the movement of air into the lungs. Air can be breathed in through the nose or through the mouth. The lining of the nose has a good supply of blood capillaries and a layer of sticky mucus.

The whole of the trachea and the larger air passages have C-shaped rings of cartilage within their walls. These rings help the air to flow freely into the lungs. The trachea divides into two tubes which then divide further into finer branches. These finer branches eventually end in air sacs. It is in the air sacs that gaseous exchange occurs.

The lungs are located in the thorax which is surrounded by the ribs. The thorax is separated from the abdomen by the diaphragm. Together, the ribs and the diaphragm can change the volume of the thorax and therefore the pressure in the thorax.

During breathing, at rest, an average of 0.50 dm^3 of air is breathed in and out. The air breathed in contains 20% oxygen and the air breathed out contains 16% oxygen. At rest, an adult will breathe 17 times a minute on average. The breathing rate is controlled by a part of the brain which receives information about carbon dioxide levels in the blood.

Carbon monoxide may be present in the air that is breathed in. The carbon monoxide can enter the blood and combine with haemoglobin in the red blood cells. Carbon monoxide is found in cigarette smoke and car exhaust fumes.

(a) (i) Discuss whether it is better to breathe in through the nose or the mouth. (lines 2 to 3)

(3)

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10. Question 2 · part c · 3 marks

January 2022 Paper 2

Further practice · Attempt part c only.

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(ii) Explain why the larger air passages have C-shaped rings of cartilage within their walls. (lines 4 to 5)

(2)

(iii) The trachea divides into two tubes.

What is the name of these tubes? (line 6)

(1)

- ☒ **A** alveoli
- ☒ **B** bronchi
- ☒ **C** bronchioles
- ☒ **D** oesophagus

(b) Which of these is the name of the part of the brain that controls breathing rate? (line 16)

(1)

- ☒ **A** cerebrum
- ☒ **B** cerebellum
- ☒ **C** hypothalamus
- ☒ **D** medulla oblongata

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10. Question 2 · part c · 3 marks

January 2022 Paper 2

Further practice · Attempt part c only.

(c) Calculate the average volume of oxygen absorbed by the lungs during one minute of breathing, at rest. (lines 13 to 15)

(3)

volume of oxygen absorbed = ... dm^3

(d) Explain why pregnant women are advised not to smoke cigarettes.

(4)

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(Total for Question 2 = 14 marks)

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10. Question 2 · part c · 3 marks · Working and answer

Working and answer

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11. Question 7 · part a · 2 marks

June 2019 Paper 2

Further practice · Attempt part a only.

7 The volume of alcohol in alcoholic drinks can be measured in units.

One unit is equivalent to 10 cm^3 of pure alcohol.

The diagram shows a bottle of wine with 14% alcohol content and a glass containing 250 cm^3 of this wine.



250 cm^3 glass of Ruby Red wine

(a) Calculate the number of units of alcohol in the glass of wine.

(2)

number of units = ...

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11. Question 7 · part a · 2 marks

June 2019 Paper 2

Further practice · Attempt part a only.

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(b) It is illegal for a person to drive a motor vehicle if they have more than the legal limit of alcohol in their bloodstream.

It takes one hour for the body to process one unit of alcohol so that there is no alcohol left in the bloodstream.

(i) A person drinks three units of alcohol. This puts her at the legal limit.

Calculate the time it would take before a person can legally drive after drinking four glasses of Ruby Red wine.

(2)

time = ... hours

(ii) Suggest why it is illegal for people to drive a motor vehicle with more than the legal limit of alcohol in their bloodstream.

(2)

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11. Question 7 · part a · 2 marks · Working and answer

Working and answer

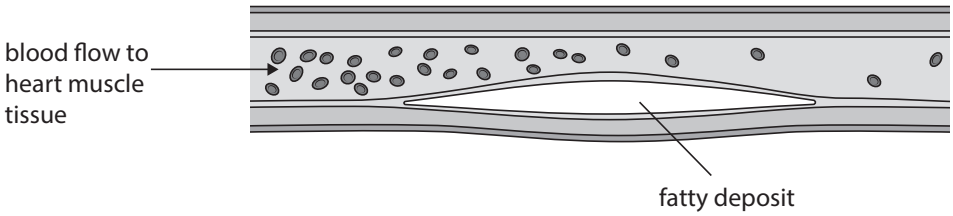
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12. Question 7 · part b.i · 3 marks

June 2024 Paper 2

Further practice · Attempt part b.i only.

7 (a) The diagram shows a fatty deposit in an artery carrying blood to the heart muscle.



(Source: adapted from https://www.heartandstroke.ca/heart-disease/conditions/atherosclerosis)

Explain how a build-up of a fatty deposit in an artery carrying blood to the heart muscle can cause heart disease.

(4)

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12. Question 7 · part b.i · 3 marks

June 2024 Paper 2

Further practice · Attempt part b.i only.

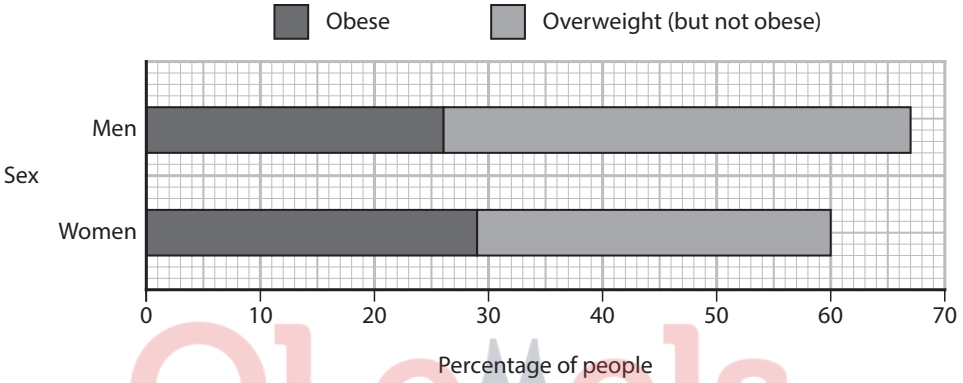
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(b) People with obesity are more at risk of heart disease than those who are overweight.

The chart shows the percentage of men and women in the United Kingdom who were either overweight or obese in 2018.



(Source: https://digital.nhs.uk/data-and-information/publications/statistical/health-survey-for-england/2018/summary)

(i) In 2018 there were 66.5 million people living in the UK. Of these people, 33 million were men.

Calculate how many more women than men were classified as obese.

(3)

extra number of women = ...

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P 7 4 6 3 0 A 0 1 9 2 4

Turn over ▶

12. Question 7 · part b.i · 3 marks

June 2024 Paper 2

Further practice · Attempt part b.i only.

(ii) Describe a calculation that could be used to determine if a person is obese.

(3)

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12. Question 7 · part b.i · 3 marks · Working and answer

Working and answer

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13. Question 3 · part b.i · 3 marks

June 2023 Paper 2

Further practice · Attempt part b.i only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Diabetes is a very common condition that affects approximately five million people in the United Kingdom. Of these five million people, one person in four has the condition but is not aware that they have it. As a result, many people are at risk of high blood sugar levels.

There are two types of diabetes, Type 1 and Type 2. Type 2 diabetes is the cause of 90% of all cases of diabetes. This type is the result of the body not producing enough insulin or the body not reacting to the insulin.

Diabetes can cause damage to blood vessels and affect the level of fluid in the body. Some people with the condition develop problems with their eyesight and hearing.

Perhaps the most common symptoms of diabetes are passing more urine, a feeling of tiredness and wounds taking longer to heal than normal.

A quick blood test can be taken to show whether someone has the condition. Early diagnosis is important as diabetes increases the risks of some complications such as strokes and heart disease.

(a) Suggest why diabetes is described as a condition rather than a disease. (line 1)

(2)

.....

.....

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.....

(b) (i) Estimate how many people in the United Kingdom have diabetes without knowing that they have it. (lines 2 and 3)

(3)

number of people = ... million

(ii) Estimate how many people in the United Kingdom have Type 1 diabetes. (lines 5 and 6)

(2)

number of people = ... million

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13. Question 3 · part b.i · 3 marks · Working and answer

Working and answer

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14. Question 3 · part b.ii · 2 marks

June 2023 Paper 2

Further practice · Attempt part b.ii only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Diabetes is a very common condition that affects approximately five million people in the United Kingdom. Of these five million people, one person in four has the condition but is not aware that they have it. As a result, many people are at risk of high blood sugar levels.

There are two types of diabetes, Type 1 and Type 2. Type 2 diabetes is the cause of 90% of all cases of diabetes. This type is the result of the body not producing enough insulin or the body not reacting to the insulin.

Diabetes can cause damage to blood vessels and affect the level of fluid in the body. Some people with the condition develop problems with their eyesight and hearing.

Perhaps the most common symptoms of diabetes are passing more urine, a feeling of tiredness and wounds taking longer to heal than normal.

A quick blood test can be taken to show whether someone has the condition. Early diagnosis is important as diabetes increases the risks of some complications such as strokes and heart disease.

(a) Suggest why diabetes is described as a condition rather than a disease. (line 1)

(2)

.....

.....

.....

.....

(b) (i) Estimate how many people in the United Kingdom have diabetes without knowing that they have it. (lines 2 and 3)

(3)

number of people = ... million

(ii) Estimate how many people in the United Kingdom have Type 1 diabetes. (lines 5 and 6)

(2)

number of people = ... million

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P 7 2 5 8 7 A 0 6 2 4

14. Question 3 · part b.ii · 2 marks

June 2023 Paper 2

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(c) (i) People with diabetes are at risk of high blood sugar levels.

State the name of the main sugar carried in blood. (line 4)

(1)

(ii) Suggest why some people with diabetes may develop problems with their eyesight and hearing. (lines 8 to 10)

(2)

(iii) Explain why passing more urine is a symptom of diabetes. (line 11)

(3)

(Total for Question 3 = 13 marks)

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14. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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